

probability

Q1 probability is the measure of uncertainty of any event, any phenomenon happens or bound to happen.

out come :- The result of any event
Like number appearing on a dice,
side of coin, drawn out card etc.

Sample space :- The set of all possible out comes, It is denoted by S .

Event :- Any combination of possible out comes or the subset of sample space. It is denoted by E .

probability,

$$p = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{n(E)}{n(S)}$$

* Total probability of any observations is $= 1$

$$* P(A \cup B) = P(A) + P(B) = P(A \text{ or } B)$$

$$* P(A \cap B) = P(A) \cdot P(B) = P(A \text{ and } B)$$

$$\{TT, HT, TH, HH\} = 2$$

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$$E = (E) \text{ or } P = (E)$$

$$24 \cdot 0 = \frac{9}{4} = 9$$

Ex:- What is the probability of getting an even number on rolling a dice once?

Soln:-

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$E = \{2, 4, 6\}$$

$$\therefore n(S) = 6 ; n(E) = 3$$

$$\therefore P = \frac{n(E)}{n(S)} = \frac{3}{6} = 0.5$$

Ex:- What is probability of getting HEAD at least once on tossing a coin twice.

Soln:-

$$S = \{HH, HT, TH, TT\}$$

$$E = \{HH, HT, TH\}$$

$$n(S) = 4 \quad n(E) = 3$$

$$\therefore P = \frac{3}{4} = 0.75$$

Ex:- what is the probability of getting all HEAD on tossing a coin three times?

Soln:-

$$S = \{ HHH, HHT, HTH, THH, HTT, THT, TTH, TTT \}$$

$$E = \{ HHH \}$$

$$n(S) = 8, n(E) = 1$$

$$\therefore P = \frac{n(E)}{n(S)}$$

$$= \frac{1}{8}$$

$$= 0.125$$

Club = চিহ্ন = ১৬

Diamond = ক্রান্ত = ১৬

Hearts = হৃৎকণ্ঠ = ১৬

Spade = ১৬

King = ৪ (২ Red, ২ black)

Queen = ৪ টি

Jack = ৪ টি

Black card = কালো

কর্তব্য = ১৬

Red card = ১৬

Ace (টাকা) = ৪ টি

Ex:- A card is drawn from a well shuffled packet. Calculate

- (i) Queen on black (iv) Black queen on black jack
 (ii) King and black (v) Red king and red
 (iii) Red queen and diamond

Soln:- Total card = 52
 Club = 13 Hearts = 13
 Diamond = 13 Black = 26
 King = 4 Red = 26
 Queen = 4
 Jack = 4

(i) Queen on black

$$P = \frac{4C_1}{52C_1} + \frac{24C_1}{52C_1} = \frac{4}{52} + \frac{24}{52} = \frac{28}{52}$$

(ii) King and black

$$P = \frac{2C_1}{52C_1} = \frac{2}{52} = \frac{1}{26}$$

(iii) Red Queen and Diamond

$$P = \frac{1C_1}{52C_1} = \frac{1}{52}$$

(iv) Red King and red

$$P = \frac{2C_1}{52C_1} = \frac{2}{52} = \frac{1}{26}$$

(v) Black queen on black jack

$$P = \frac{2C_1}{52C_1} + \frac{2C_1}{52C_1} = \frac{2}{52} + \frac{2}{52} = \frac{4}{52} = \frac{1}{13}$$

Ex: There are 2 green, 3 black and 5 yellow balls in a bag. If two balls are drawn at random, what will be the probability?

(i) 2 yellow

(ii) 1 black

(iii) 1 green and 1 yellow

(iv) 2 green or 1 black

Soln:-

$$\text{Total ball } n(s) = 2 + 3 + 5 = 10$$

$$(i) P(2Y) = \frac{{}^5C_2}{{}^{10}C_2} = 0.22$$

$$(ii) P(1B) = \frac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2} = 0.47$$

$$(iii) P(1G \text{ and } 1Y) = \frac{{}^2C_1 \times {}^5C_1}{{}^{10}C_2} = 0.22$$

$$(iv) P(2G \text{ and } 1 \text{ black}) = \frac{{}^2C_2}{{}^{10}C_2} + \frac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2} = 0.49$$

Ex: A bag contains 4 red and 6 black balls. Two balls are drawn from the bag at random. Find the following:

- (i) The balls are black.
- (ii) The balls are same colour.
- (iii) The balls are of different colour.

Soln:-

$$\text{Red balls} = 4$$

$$\text{Black balls} = 6$$

$$\text{Total balls} = 4 + 6 = 10$$

Two balls are drawn at random

$$(i) \quad P = \frac{{}^6C_2}{{}^{10}C_2} = \frac{15}{45}$$

$$(ii) \quad P = \left(\frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^4C_2}{{}^{10}C_2} \right) \\ = \frac{15}{45} + \frac{6}{45} = \frac{21}{45}$$

$$\begin{aligned}
 \text{(c)} \quad P &= \frac{{}^4C_1 \times {}^6C_1}{{}^{10}C_2} \\
 &= \frac{4 \times 6}{45} \\
 &= \frac{24}{45}
 \end{aligned}$$

Ex:- A pot contains 7 white and 3 black balls, another pot contains 5 white and 7 black balls. A pot is selected at random and 2 balls are drawn from it without replacement. Find the following probabilities :- (a) two balls are white and (b) two balls are same in color.

Soln:-

In 1st pot,

white balls = 7

black balls = 3

$\therefore$ Total balls = 10

In 2nd pot,

white balls = 5

black balls = 7

$\therefore$ Total balls = 12

(a) The probability of getting two balls are white.

$$\begin{aligned}
 P &= \frac{1}{2} \left(\frac{{}^7C_1}{{}^{10}C_1} \times \frac{{}^6C_1}{{}^9C_1} \right) + \frac{1}{2} \left(\frac{{}^5C_1}{{}^{12}C_1} \times \frac{{}^4C_1}{{}^{11}C_1} \right) \\
 &= 0.030
 \end{aligned}$$

(b)

The probability of getting black,

$$P = \frac{1}{2} \left(\frac{3C_1}{10C_1} \times \frac{2C_1}{9C_1} \right) + \frac{1}{2} \left(\frac{7C_1}{12C_1} \times \frac{6C_1}{11C_1} \right)$$
$$= 0.1924$$

∴ The probability of getting same color,

$$P = 0.030 + 0.1924$$
$$= 0.2224$$

Ans:-

Ex:- A bag contains 7 red and 3 black balls. Another bag contains 6 red and 6 black balls. A ball is taken from the first bag and put in the second bag. Then two balls are drawn at random from the second bag. What is the probability that the two balls will be same color?

Soln:- 1st bag -
Red = 7
Black = 3
Total = 10

2nd bag.
Red = 6
Black = 6
Total = 12

R+R
R+B
B+R
B+B

The required probability :-

$$p = \left(\frac{7c_1}{10c_1} \times \frac{7c_2}{13c_2} \right) + \left(\frac{7c_1}{10c_1} \times \frac{6c_2}{13c_2} \right) + \left(\frac{3c_1}{10c_1} \times \frac{6c_2}{13c_2} \right) + \left(\frac{3c_1}{10c_1} \times \frac{7c_2}{13c_2} \right)$$
$$= \left(\frac{7}{10} \times \frac{21}{78} \right) + \left(\frac{7}{10} \times \frac{15}{78} \right) + \left(\frac{3}{10} \times \frac{15}{78} \right) + \left(\frac{3}{10} \times \frac{21}{78} \right)$$
$$= 0.461538$$

Ans: —

Ex:- 3 cards are drawn at random from a packet of 52 card. Find the probability of getting 1. Two King. 2. Two hearts 3. 1 spade 1 club and 1 diamond.

Soln:-

Total card = 52

King = 4

Hearts = 13

club = 13

Diamond = 13

3 cards are drawn from a packet.

1. The probability of getting two

King :

$$p = \frac{{}^4C_2 \times {}^{48}C_1}{{}^{52}C_3} = \frac{6 \times 48}{22100} = \frac{72}{5525}$$

2. Two hearts :

$$p = \frac{{}^{13}C_2 \times {}^{39}C_1}{{}^{52}C_3} = \frac{78 \times 39}{22100} = \frac{117}{850}$$

3. The probability of getting 1 spade,
1 club and 1 diamond.

$$p = \frac{{}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1}{{}^{52}C_3} = \frac{13 \times 13 \times 13}{22100}$$

$$= \frac{169}{1700}$$

Ans: -